

Independent-model topside retrieval: NeQuick-G validation

29 September 2026

Summary

A vertical, mode-resolved synthetic ionogram was generated from a public **NeQuick-G** density profile that was not included in the retrieval basis. The April validation profile was chosen at 62°S, 8°W, 13 UT with Galileo broadcast Az = 64. The profile was replicated horizontally over the existing ray grid. After full O/X ray checks of eight candidates, the minimum-score retrieval had **−0.032 MHz foF2 error, −13.9 km hmF2 error, and −1.1% peak density error**. Its peak-normalized topside RMS was **7.9%** from the true F2 peak to 600 km. At 800 km it still underestimated density by **42.5%**. The selected O/X ionogram score was 0.181; smaller is better.

This is a useful out-of-input-family check, but it does not establish robust recovery of the upper topside. The 2–10 MHz sweep contains little direct information about plasma whose local frequency is below 2 MHz. Several different upper tails therefore give plausible ionograms.

Truth and independence

The truth generator used the public `tpl2go/NequickG` implementation at commit `1d1783412ed9` (full hash in `provenance.json`). Its Python 2 source was converted mechanically with `lib2to3`; the model equations were not edited. The retrieval’s eight-slope, monotone log-density spline contains **no NeQuick profiles, coefficients, or source code**. The fitting script reads only the saved O/X returns. It wrote `selection.json` with all candidate scores and the selected name **before** opening the truth density or NeQuick peak parameters. The earlier January and July NeQuick cases informed method development; this April location and time were held out from density-based assessment until that selection.

All truth and candidate ionograms use the same 3-D PyLap ray tracer, option-D equal-area vertical fan, 2–10 MHz frequencies at 100 kHz spacing, and 1 km homing gate. The forward physics is therefore shared. This experiment tests independence of the **density model**, not independence of the ray tracer. Replicating one NeQuick profile over the horizontal grid also removes real horizontal structure; it is a controlled vertical-profile test, not a global NeQuick simulation or a measured ionogram.

Retrieval and selection

The proposal fit represents the density above hmF2 with eight positive piecewise log-density slopes. foF2, hmF2, and those slopes are fitted to the median O-mode group range at each accepted frequency using a fast scalar vertical-ray integral. A weak penalty discourages sharp slope changes. The tail-regularized branch enforces a minimum positive gradient in the upper three intervals to avoid an unphysical plateau. Its 55 km Chapman bottomside is a fixed nuisance shape. The scalar fit is only a proposal: it omits X-mode magnetoionic bending and is not the final score.

The candidate set also tested foF2 anchors inferred from the X-mode last return, subject to the lower bound supplied by the O-mode last return. Three additional candidates imposed generic 0.4, 0.8, or 1.2 MHz plasma-frequency anchors at the 800 km spacecraft. These were **assumed tail priors**, not measured plasma frequencies. Every candidate was then traced with full O/X rays and ranked by the same accepted-return score: 55% symmetric return distance, 25% mode-wise last-return frequency, and 20% common-frequency median range error. All accepted rays enter the score.

Ionograms and density cut

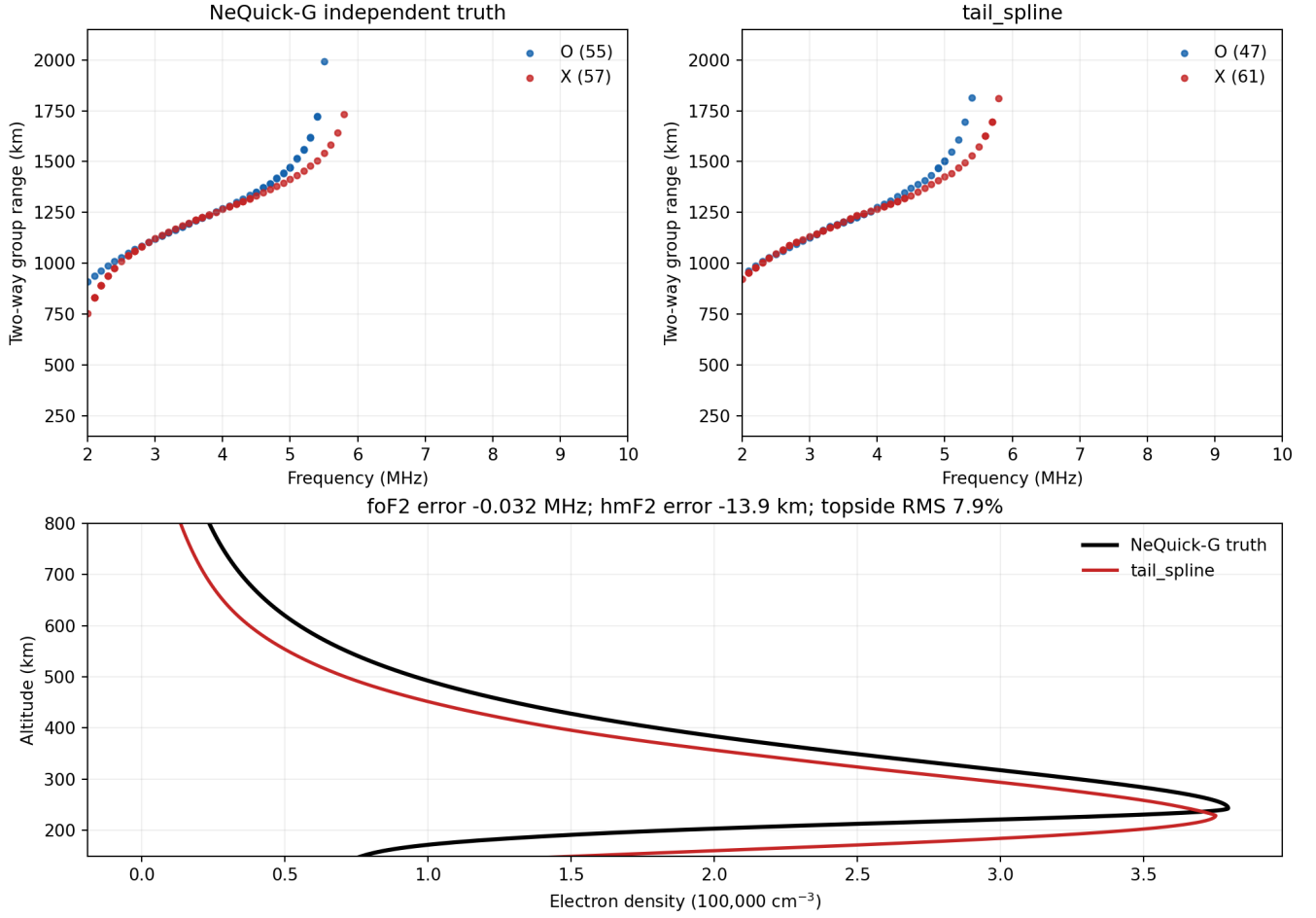


Figure 1: April NeQuick-G truth and score-selected retrieval. O returns are blue and X returns red in each ionogram; every accepted return is plotted. The lower panel compares the independent density truth with the retrieved vertical cut. The near-peak ionograms agree substantially better than the upper density tails.

The score-selected spline follows the low and middle frequency return ridges and places foF2 within 32 kHz of the model value. It retrieves a lower peak height and a steeper topside. The large remaining density difference above 500 km is visible even though its contribution to peak-normalized RMS is small.

Quantitative result

The April NeQuick-G model has foF2 **5.532 MHz** and hmF2 **243.2 km**. The table reports density errors only after the ionogram-score selection. Topside RMS is normalized by the true peak density and covers the true peak through 600 km. One mode-frequency bin means one 100 kHz frequency in one polarization.

Candidate	O/X score ↓	foF2 Δ (MHz)	hmF2 Δ (km)	Peak Δ	Top RMS
Legacy O-nose spline	0.182	−0.032	−13.9	−1.1%	7.9%
Selected tail-regularized spline	0.181	−0.032	−13.9	−1.1%	7.9%
X-anchor low	0.233	−0.024	−44.2	−0.9%	20.3%
X-anchor mid	0.228	+0.120	−27.8	+4.4%	7.3%
X-anchor high	0.243	+0.216	−49.1	+8.0%	16.8%
0.4 MHz upper anchor	0.217	+0.119	−56.2	+4.3%	19.7%
0.8 MHz upper anchor	0.241	+0.120	−35.6	+4.4%	10.4%
1.2 MHz upper anchor ¹	0.241	+0.120	−23.4	+4.4%	5.6%

¹ Four interior mode-frequency bins remained unhomed after the candidate’s gap search, so its score has that search limitation. Its score is still 0.060 above the selected candidate’s score.

The selected density is 13,548 cm^{−3} at 800 km, versus 23,567 cm^{−3} in NeQuick-G. The relative bias grows from −11.5% at 300 km to −19.4% at 400 km, −31.8% at 600 km, and −42.5% at 800 km. This is the unresolved topside-shape error; a small peak-density error should not be read as a good upper-tail retrieval.

Interpretation

The flexible spline can represent density shapes outside the IRI or Chapman input families, and the held-out April case recovers the peak reasonably well. The full-ray score does **not** reliably choose the correct upper tail from this 2 MHz lower sweep limit. Extending observations below 2 MHz or adding an independent upper-plasma constraint would test and reduce that ambiguity. Further tuning on this April truth would make it a development case; a new held-out profile would then be needed for a prospective check.

Forward-model quality control

Interior frequency gaps in the truth and candidates were searched with full-fan seeds and 20 kHz continuation. Only rays actually traced and homed at their displayed 100 kHz frequency were added. The April truth gained 14 accepted returns and had no remaining interior gaps. Seven candidate gap passes closed their gaps; one unselected candidate retained four unrecovered mode-frequency bins. Jobs ran in `chartat1`'s private Cartman run area, with at most two large ray processes active.

The earlier January NeQuick truth sweep had a five-bin X-mode gap from 2.5 through 2.9 MHz. There were returns at 2.4 and 3.0 MHz. A dense fan found rays at 2.6–2.8 MHz; 20 kHz continuation from 2.4, 2.6, 2.8, and 3.0 MHz found two accepted X rays at **each** missing 100 kHz frequency. All ten passed the original 1 km gate; the largest miss was 780 m. Figure 2 shows the original and corrected truth ionograms, including a close view. The previous January retrieval score table used the incomplete truth and remains preliminary until its candidates receive the same recovery pass.

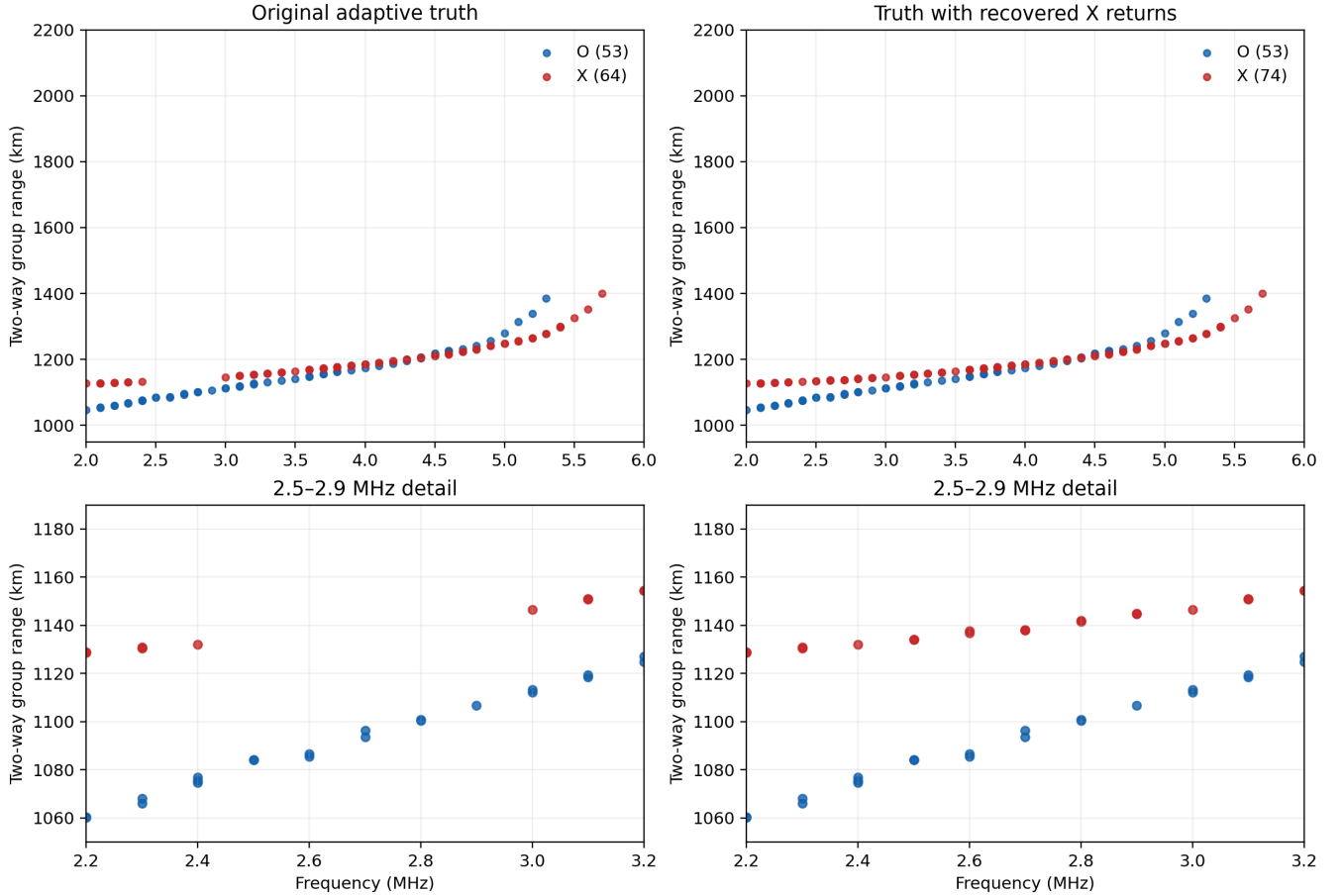


Figure 2: Original January NeQuick-G truth (left) and the exact-frequency gap-recovered truth (right). Blue is O mode and red is X mode. The lower panels resolve the 2.5–2.9 MHz interval; all accepted returns are retained.